Stabilized Low-Rank Quaternion Matrix Completion with Complementary Local and Nonlocal Plug-and-Play Priors for Color Image Inpainting (Supplementary Material)

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Appendix A. Proof of Theorem 3.1

Theorem 3.1 Let $\dot{\mathbf{A}} \in \mathbb{H}^{m\times n}$ have the QSVD $\dot{\mathbf{A}} = \dot{\mathbf{U}}\dot{\mathbf{\Sigma}}\dot{\mathbf{V}}^H$, with singular values $\sigma_1 \geq \cdots \geq \sigma_r \geq 0$, where $r = \min\{m, n\}$. For $\tau > 0$ and $\alpha > 0$, consider

$$\min_{\dot{\mathbf{W}}} \frac{1}{2}\|\dot{\mathbf{W}} - \dot{\mathbf{A}}\|_F^2 + \tau\left(\|\dot{\mathbf{W}}\|_* - \alpha\|\dot{\mathbf{W}}\|_F\right). \tag{A.1}$$

Define $s_i = (\sigma_i - \tau)_+$, $\mathbf{s} = (s_1, \ldots, s_r)^T$, where $(t)_+ = \max\{t, 0\}$, and let $\mathbf{e}_1 = (1, 0, \ldots, 0)^T$. Then, a global minimizer of (A.1) is given by $\dot{\mathbf{W}}^\star = \dot{\mathbf{U}}\,\mathrm{diag}(\widehat{\boldsymbol{\sigma}})\dot{\mathbf{V}}^H$, where

$$\widehat{\boldsymbol{\sigma}} = \begin{cases} \left(1 + \dfrac{\alpha\tau}{\|\mathbf{s}\|_2}\right)\mathbf{s}, & \sigma_1 > \tau, \\ [\sigma_1 - (1-\alpha)\tau]\,\mathbf{e}_1, & (1-\alpha)\tau < \sigma_1 \leq \tau, \\ \mathbf{0}, & \sigma_1 \leq (1-\alpha)\tau. \end{cases} \tag{A.2}$$

Proof. Let $\dot{\mathbf{W}} = \widetilde{\mathbf{U}}\,\mathrm{diag}(\mathbf{w})\widetilde{\mathbf{V}}^H$ be a QSVD of $\dot{\mathbf{W}}$, where $\mathbf{w} = (w_1, \ldots, w_r)^T$ contains its nonnegative singular values.

By the quaternion von Neumann trace inequality,

$$\mathrm{Re}\,\mathrm{tr}(\dot{\mathbf{W}}^H\dot{\mathbf{A}}) \leq \sum_{i=1}^{r} w_i\sigma_i.$$

Equality is attained when the singular vectors of $\dot{\mathbf{W}}$ are aligned with those of $\dot{\mathbf{A}}$. Therefore, it suffices to solve

$$\min_{\mathbf{w}\geq\mathbf{0}} \frac{1}{2}\|\mathbf{w} - \boldsymbol{\sigma}\|_2^2 + \tau\left(\|\mathbf{w}\|_1 - \alpha\|\mathbf{w}\|_2\right), \tag{A.3}$$

where $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_r)^T$. The solution constructed below also preserves the nonincreasing order of the singular values.

Define $\mathbf{d} = \boldsymbol{\sigma} - \tau\mathbf{1}$. Expanding the objective in (A.3) gives

$$F(\mathbf{w}) = \frac{1}{2}\|\boldsymbol{\sigma}\|_2^2 + \frac{1}{2}\|\mathbf{w}\|_2^2 - \mathbf{d}^T\mathbf{w} - \alpha\tau\|\mathbf{w}\|_2.$$

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For any nonzero $\mathbf{w} \geq \mathbf{0}$, write $\mathbf{w} = t\mathbf{u}$, where $t > 0$, $\mathbf{u} \geq \mathbf{0}$, and $\|\mathbf{u}\|_2 = 1$. Then

$$F(t\mathbf{u}) = \frac{1}{2}\|\sigma\|_2^2 + \frac{1}{2}t^2 - t\left(\mathbf{d}^T\mathbf{u} + \alpha\tau\right). \tag{A.4}$$

For a fixed direction $\mathbf{u}$, the optimal radius is

$$t^\star(\mathbf{u}) = \left(\mathbf{d}^T\mathbf{u} + \alpha\tau\right)_+. \tag{A.5}$$

It remains to maximize $\mathbf{d}^T\mathbf{u}$ over $\mathbf{u} \geq \mathbf{0}$ with $\|\mathbf{u}\|_2 = 1$.

Case 1: $\sigma_1 > \tau$. In this case, $\mathbf{s} = (\mathbf{d})_+ \neq \mathbf{0}$. By the Cauchy–Schwarz inequality,

$$\mathbf{d}^T\mathbf{u} \leq \mathbf{s}^T\mathbf{u} \leq \|\mathbf{s}\|_2,$$

and equality is attained at

$$\mathbf{u}^\star = \frac{\mathbf{s}}{\|\mathbf{s}\|_2}.$$

It follows from (A.5) that

$$t^\star = \|\mathbf{s}\|_2 + \alpha\tau.$$

Consequently,

$$\widehat{\boldsymbol{\sigma}} = \left(1 + \frac{\alpha\tau}{\|\mathbf{s}\|_2}\right)\mathbf{s}.$$

Case 2: $\sigma_1 \leq \tau$. Here, $d_i \leq 0$ for every i. Since $\mathbf{u} \geq \mathbf{0}$ and $\|\mathbf{u}\|_2 = 1$, we have $\|\mathbf{u}\|_1 \geq 1$ and

$$\mathbf{d}^T\mathbf{u} \leq d_1\|\mathbf{u}\|_1 \leq d_1 = \sigma_1 - \tau.$$

This upper bound is attained by $\mathbf{u}^\star = \mathbf{e}_1$. Therefore,

$$t^\star = [\sigma_1 - (1-\alpha)\tau]_+.$$

If $(1-\alpha)\tau < \sigma_1 \leq \tau$, then $t^\star > 0$ and

$$\widehat{\boldsymbol{\sigma}} = [\sigma_1 - (1-\alpha)\tau]\,\mathbf{e}_1.$$

Otherwise, $t^\star = 0$ and $\widehat{\boldsymbol{\sigma}} = \mathbf{0}$.

Combining the two cases establishes (A.2). Since the corresponding singular values are nonnegative and nonincreasing, and alignment of the singular vectors attains the quaternion trace bound, the resulting matrix $\dot{\mathbf{W}}^\star$ is a global minimizer of (A.1). □

Appendix B. Proof of Theorem 3.2

Theorem 3.2 Under Assumption 1, let $\{(\dot{\mathbf{X}}^k, \dot{\mathbf{Z}}^k, \mathcal{S}^k, \mathcal{T}^k)\}_{k=0}^\infty$ be the sequence of primal variables generated by SGLN-LRQMC. Then $\{\dot{\mathbf{X}}^k\}_{k=0}^\infty$, $\{\dot{\mathbf{Z}}^k\}_{k=0}^\infty$, $\{\mathcal{S}^k\}_{k=0}^\infty$, and $\{\mathcal{T}^k\}_{k=0}^\infty$ are Cauchy sequences, and hence there exist $\dot{\mathbf{X}}^\star$, $\dot{\mathbf{Z}}^\star$, $\mathcal{S}^\star$, and $\mathcal{T}^\star$ such that

$$\dot{\mathbf{X}}^k \to \dot{\mathbf{X}}^\star, \quad \dot{\mathbf{Z}}^k \to \dot{\mathbf{Z}}^\star, \quad \mathcal{S}^k \to \mathcal{S}^\star, \quad \mathcal{T}^k \to \mathcal{T}^\star.$$

Proof. To establish the convergence of the primal iterates, it suffices to prove the finite-length property

$$\sum_{k=0}^{\infty} \left\|\dot{\mathbf{X}}^{k+1} - \dot{\mathbf{X}}^{k}\right\|_F < \infty \tag{B.1}$$

and show that the following three primal residuals vanish:

$$\lim_{k\to\infty} \left\|\dot{\mathbf{X}}^{k+1} - \dot{\mathbf{Z}}^{k+1}\right\|_F = 0, \tag{B.2}$$

$$\lim_{k\to\infty} \left\|\Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) - \mathcal{S}^{k+1}\right\|_F = 0, \tag{B.3}$$

$$\lim_{k\to\infty} \left\|\Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) - \mathcal{T}^{k+1}\right\|_F = 0. \tag{B.4}$$

Indeed, (B.1) implies that $\{\dot{\mathbf{X}}^k\}$ is a Cauchy sequence. The vanishing primal residuals then imply that $\{\dot{\mathbf{Z}}^k\}$, $\{\mathcal{S}^k\}$, and $\{\mathcal{T}^k\}$ converge as well.

Since $\mu^{k+1} = \rho\mu^k$ with $\mu^0 > 0$ and $\rho > 1$,

$$\mu^k = \mu^0\rho^k \to \infty. \tag{B.5}$$

We first derive bounds associated with the two PnP updates. For the local denoising branch,

$$\mathcal{S}^{k+1} = T_{g_L^k}^{-1}\mathfrak{D}^L\left(T_{g_L^k}\left(\Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) + \frac{\mathcal{U}^k}{\mu^k}\right), \sqrt{\frac{\gamma}{\mu^k}}\right). \tag{B.6}$$

Since $T_g^{-1}T_g = I$ and every transformation $T_g \in \mathcal{G}$ preserves the Frobenius norm, Assumption 1 yields

$$\begin{aligned}
&\left\|\mathcal{S}^{k+1} - \Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) - \frac{\mathcal{U}^k}{\mu^k}\right\|_F \\
&= \left\|\mathfrak{D}^L\left(T_{g_L^k}\left(\Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) + \frac{\mathcal{U}^k}{\mu^k}\right), \sqrt{\frac{\gamma}{\mu^k}}\right) - T_{g_L^k}\left(\Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) + \frac{\mathcal{U}^k}{\mu^k}\right)\right\|_F \\
&\le \sqrt{3mnC}\sqrt{\frac{\gamma}{\mu^k}} = \frac{c_L}{\sqrt{\mu^k}},
\end{aligned} \tag{B.7}$$

where $c_L = \sqrt{3mnC\gamma}$.

Similarly, the nonlocal denoising update satisfies

$$\begin{aligned}
&\left\|\mathcal{T}^{k+1} - \Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) - \frac{\mathcal{V}^k}{\mu^k}\right\|_F \\
&\le \sqrt{3mnC}\sqrt{\frac{\eta}{\mu^k}} = \frac{c_N}{\sqrt{\mu^k}},
\end{aligned} \tag{B.8}$$

where $c_N = \sqrt{3mnC\eta}$.

Consequently,

$$\begin{aligned}
\left\|\mathcal{S}^{k+1} - \Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) - \frac{\mathcal{U}^k}{\mu^k}\right\|_F &\le \frac{c_L}{\sqrt{\mu^k}}, \\
\left\|\mathcal{T}^{k+1} - \Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) - \frac{\mathcal{V}^k}{\mu^k}\right\|_F &\le \frac{c_N}{\sqrt{\mu^k}}.
\end{aligned} \tag{B.9}$$

By the multiplier updates,

$$\mathcal{U}^{k+1} = \mathcal{U}^k + \mu^k \left(\Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) - \mathcal{S}^{k+1}\right),$$
$$\mathcal{V}^{k+1} = \mathcal{V}^k + \mu^k \left(\Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) - \mathcal{T}^{k+1}\right),$$

we obtain

$$\|\mathcal{U}^{k+1}\|_F \le c_L \sqrt{\mu^k}, \qquad \|\mathcal{V}^{k+1}\|_F \le c_N \sqrt{\mu^k}. \tag{B.10}$$

Next, recall that

$$\dot{\mathbf{E}}^k = \dot{\mathbf{Z}}^{k+1} - \frac{\dot{\mathbf{\Lambda}}^k}{\mu^k}, \quad \mathcal{M}^k = \mathcal{S}^k - \frac{\mathcal{U}^k}{\mu^k}, \quad \mathcal{N}^k = \mathcal{T}^k - \frac{\mathcal{V}^k}{\mu^k}.$$

Let

$$\dot{\mathbf{X}}^k + \frac{\dot{\mathbf{\Lambda}}^k}{\mu^k} = \dot{\mathbf{U}}^k \operatorname{diag}(\sigma_1^k, \ldots, \sigma_r^k)(\dot{\mathbf{V}}^k)^H \tag{B.11}$$

be its QSVD, where $r = \min\{m, n\}$.

By the QNMF shrinkage formula, for $0 < \alpha \le 1$, the singular values $\widehat{\sigma}_i^k$ of $\dot{\mathbf{Z}}^{k+1}$ satisfy

$$\left|\widehat{\sigma}_i^k - \sigma_i^k\right| \le \frac{\max\{1, \alpha\}\lambda}{\mu^k}, \qquad i = 1, \ldots, r. \tag{B.12}$$

Let $d_G = \max\{1, \alpha\}\lambda\sqrt{r}$. Since the QNMF update uses the singular vectors in (B.11), the unitary invariance of the Frobenius norm gives

$$\begin{aligned} &\left\|\dot{\mathbf{E}}^k - \dot{\mathbf{X}}^k\right\|_F^2 \\ &= \left\|\dot{\mathbf{Z}}^{k+1} - \left(\dot{\mathbf{X}}^k + \frac{\dot{\mathbf{\Lambda}}^k}{\mu^k}\right)\right\|_F^2 \\ &= \sum_{i=1}^{r} \left(\widehat{\sigma}_i^k - \sigma_i^k\right)^2 \le \frac{d_G^2}{(\mu^k)^2}. \end{aligned} \tag{B.13}$$

Hence,

$$\left\|\dot{\mathbf{E}}^k - \dot{\mathbf{X}}^k\right\|_F \le \frac{d_G}{\mu^k}. \tag{B.14}$$

For $k \ge 2$, using (B.9), (B.10), and $\mu^k = \rho\mu^{k-1}$, we have

$$\begin{aligned} &\left\|\mathcal{M}^k - \Gamma^{-1}(\dot{\mathbf{X}}^k)\right\|_F \\ &\le \left\|\mathcal{S}^k - \Gamma^{-1}(\dot{\mathbf{X}}^k) - \frac{\mathcal{U}^{k-1}}{\mu^{k-1}}\right\|_F + \frac{\|\mathcal{U}^{k-1}\|_F}{\mu^{k-1}} + \frac{\|\mathcal{U}^k\|_F}{\mu^k} \\ &\le \frac{c_L}{\sqrt{\mu^{k-1}}} + \frac{c_L\sqrt{\mu^{k-2}}}{\mu^{k-1}} + \frac{c_L\sqrt{\mu^{k-1}}}{\mu^k} \\ &= \frac{d_L}{\sqrt{\mu^{k-1}}}, \end{aligned} \tag{B.15}$$

where

$$d_L = c_L(1 + \rho^{-1/2} + \rho^{-1}).$$

The same argument gives

$$\left\|\mathcal{N}^k - \Gamma^{-1}(\dot{\mathbf{X}}^k)\right\|_F \le \frac{d_N}{\sqrt{\mu^{k-1}}}, \qquad k \ge 2, \tag{B.16}$$

where

$$d_N = c_N(1 + \rho^{-1/2} + \rho^{-1}).$$

Define

$$\dot{\mathbf{R}}^k = \dot{\mathbf{E}}^k - \dot{\mathbf{X}}^k + \Gamma\left(\mathcal{M}^k - \Gamma^{-1}(\dot{\mathbf{X}}^k)\right) + \Gamma\left(\mathcal{N}^k - \Gamma^{-1}(\dot{\mathbf{X}}^k)\right). \tag{B.17}$$

Since Γ preserves the Frobenius norm, the preceding estimates imply

$$\begin{aligned} \|\dot{\mathbf{R}}^k\|_F &\le \frac{d_G}{\mu^k} + \frac{d_L + d_N}{\sqrt{\mu^{k-1}}} \\ &\le \frac{d}{\sqrt{\mu^{k-1}}}, \qquad k \ge 2, \end{aligned} \tag{B.18}$$

for some constant $d > 0$ independent of k.

We now derive a bound for $\dot{\mathbf{X}}^{k+1} - \dot{\mathbf{X}}^k$. Recall that $\mathcal{R}$ and $\mathcal{I}$ denote the real part and pure imaginary part projections, respectively, and

$$\dot{\mathbf{B}}_I^k = \frac{\mathcal{I}(\dot{\mathbf{E}}^k) + \Gamma(\mathcal{M}^k) + \Gamma(\mathcal{N}^k)}{3}.$$

Since $\Gamma(\mathcal{M}^k)$ and $\Gamma(\mathcal{N}^k)$ are pure quaternion matrices, (B.17) gives

$$\mathcal{R}(\dot{\mathbf{E}}^k) = \mathcal{R}(\dot{\mathbf{X}}^k) + \mathcal{R}(\dot{\mathbf{R}}^k)$$

and

$$\dot{\mathbf{B}}_I^k = \mathcal{I}(\dot{\mathbf{X}}^k) + \frac{1}{3}\mathcal{I}(\dot{\mathbf{R}}^k).$$

Substituting these identities into the explicit $\dot{\mathbf{X}}$ update and subtracting $\dot{\mathbf{X}}^k$, we obtain

$$\dot{\mathbf{X}}^{k+1} - \dot{\mathbf{X}}^k = \mathcal{P}_{\Omega^c}\left(\mathcal{R}(\dot{\mathbf{R}}^k) + \frac{1}{3}\mathcal{I}(\dot{\mathbf{R}}^k)\right) + \mathcal{P}_{\Omega}\left(\frac{\mu^k\mathcal{R}(\dot{\mathbf{R}}^k) + \mathcal{R}(\dot{\mathbf{Y}} - \dot{\mathbf{X}}^k)}{1+\mu^k} + \frac{\mu^k\mathcal{I}(\dot{\mathbf{R}}^k) + \mathcal{I}(\dot{\mathbf{Y}} - \dot{\mathbf{X}}^k)}{1+3\mu^k}\right). \tag{B.19}$$

The real and pure imaginary projections are orthogonal under the Frobenius inner product. Moreover, $\mathcal{P}_\Omega$ and $\mathcal{P}_{\Omega^c}$ are orthogonal coordinate projections and commute with $\mathcal{R}$ and $\mathcal{I}$. Since

$$\frac{\mu^k}{1+\mu^k} \le 1, \qquad \frac{\mu^k}{1+3\mu^k} \le 1, \qquad \frac{1}{1+3\mu^k} \le \frac{1}{1+\mu^k},$$

the triangle inequality and the nonexpansiveness of these projections yield

$$\begin{aligned}
&\left\|\dot{\mathbf{X}}^{k+1}-\dot{\mathbf{X}}^{k}\right\|_F \\
&\le \|\dot{\mathbf{R}}^{k}\|_F+\frac{\|\dot{\mathbf{Y}}-\dot{\mathbf{X}}^{k}\|_F}{1+\mu^{k}} \\
&\le \frac{d}{\sqrt{\mu^{k-1}}}+\frac{\|\dot{\mathbf{Y}}\|_F+\|\dot{\mathbf{X}}^{k}\|_F}{1+\mu^{k}}, \qquad k\ge 2.
\end{aligned} \tag{B.20}$$

Consequently,

$$\|\dot{\mathbf{X}}^{k+1}\|_F \le (1+a_k)\|\dot{\mathbf{X}}^{k}\|_F + b_k, \qquad k\ge 2, \tag{B.21}$$

where

$$a_k=\frac{1}{1+\mu^{k}}, \qquad b_k=\frac{d}{\sqrt{\mu^{k-1}}}+\frac{\|\dot{\mathbf{Y}}\|_F}{1+\mu^{k}}.$$

Since $\mu^k=\mu^0\rho^k$ with $\rho>1$, both series

$$\sum_{k=2}^{\infty} a_k \quad \text{and} \quad \sum_{k=2}^{\infty} b_k$$

are convergent. Iterating (B.21) and using $1+t\le e^t$ for $t\ge 0$, we obtain

$$\begin{aligned}
\|\dot{\mathbf{X}}^{k+1}\|_F &\le \left(\|\dot{\mathbf{X}}^{2}\|_F+\sum_{j=2}^{k} b_j\right)\prod_{j=2}^{k}(1+a_j) \\
&\le \left(\|\dot{\mathbf{X}}^{2}\|_F+\sum_{j=2}^{\infty} b_j\right)\exp\left(\sum_{j=2}^{\infty} a_j\right)<\infty.
\end{aligned} \tag{B.22}$$

Thus, $\{\dot{\mathbf{X}}^k\}$ is bounded. Let $M_X>0$ satisfy

$$\|\dot{\mathbf{X}}^{k}\|_F \le M_X, \qquad k\ge 0.$$

Combining this bound with (B.20) gives

$$\left\|\dot{\mathbf{X}}^{k+1}-\dot{\mathbf{X}}^{k}\right\|_F \le \frac{d}{\sqrt{\mu^{k-1}}}+\frac{\|\dot{\mathbf{Y}}\|_F+M_X}{1+\mu^{k}}, \qquad k\ge 2. \tag{B.23}$$

Since the penalty sequence grows geometrically,

$$\sum_{k=2}^{\infty}\frac{1}{\sqrt{\mu^{k-1}}}<\infty, \qquad \sum_{k=2}^{\infty}\frac{1}{1+\mu^{k}}<\infty. \tag{B.24}$$

It follows that

$$\sum_{k=2}^{\infty}\left\|\dot{\mathbf{X}}^{k+1}-\dot{\mathbf{X}}^{k}\right\|_F<\infty. \tag{B.25}$$

Adding the finitely many initial terms establishes (B.1).

For any integers $p > q \geq 2$, the triangle inequality gives

$$\begin{aligned} \left\|\dot{\mathbf{X}}^p - \dot{\mathbf{X}}^q\right\|_F &\leq \sum_{k=q}^{p-1} \left\|\dot{\mathbf{X}}^{k+1} - \dot{\mathbf{X}}^k\right\|_F \\ &\leq \sum_{k=q}^{\infty} \left\|\dot{\mathbf{X}}^{k+1} - \dot{\mathbf{X}}^k\right\|_F . \end{aligned} \tag{B.26}$$

The final term is the tail of a convergent series and therefore tends to zero as $q \to \infty$. Hence, $\{\dot{\mathbf{X}}^k\}$ is a Cauchy sequence. Since $\mathbb{H}^{m\times n}$ is complete under the Frobenius norm, there exists $\dot{\mathbf{X}}^\star \in \mathbb{H}^{m\times n}$ such that

$$\dot{\mathbf{X}}^k \to \dot{\mathbf{X}}^\star. \tag{B.27}$$

In particular,

$$\lim_{k\to\infty} \left\|\dot{\mathbf{X}}^{k+1} - \dot{\mathbf{X}}^k\right\|_F = 0. \tag{B.28}$$

It remains to establish the vanishing of the three primal residuals. By the multiplier update

$$\dot{\mathbf{\Lambda}}^{k+1} = \dot{\mathbf{\Lambda}}^{k} + \mu^k \left(\dot{\mathbf{X}}^{k+1} - \dot{\mathbf{Z}}^{k+1}\right)$$

and the definition of $\dot{\mathbf{E}}^k$, we have

$$\begin{aligned} \frac{\dot{\mathbf{\Lambda}}^{k+1}}{\mu^{k+1}} &= \frac{1}{\rho}\left(\dot{\mathbf{X}}^{k+1} - \dot{\mathbf{E}}^k\right) \\ &= \frac{1}{\rho}\left[\dot{\mathbf{X}}^{k+1} - \dot{\mathbf{X}}^k - (\dot{\mathbf{E}}^k - \dot{\mathbf{X}}^k)\right] \to 0, \end{aligned} \tag{B.29}$$

where the limit follows from (B.14) and (B.28).

Consequently,

$$\begin{aligned} &\left\|\dot{\mathbf{X}}^{k+1} - \dot{\mathbf{Z}}^{k+1}\right\|_F \\ &= \left\|\frac{\dot{\mathbf{\Lambda}}^{k+1} - \dot{\mathbf{\Lambda}}^{k}}{\mu^k}\right\|_F \\ &\leq \rho\frac{\|\dot{\mathbf{\Lambda}}^{k+1}\|_F}{\mu^{k+1}} + \frac{\|\dot{\mathbf{\Lambda}}^{k}\|_F}{\mu^k} \to 0. \end{aligned} \tag{B.30}$$

Thus, (B.2) holds.

Finally, by the multiplier updates and (B.10),

$$\begin{aligned} &\left\|\Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) - \mathcal{S}^{k+1}\right\|_F \\ &\leq \frac{\|\mathcal{U}^{k+1}\|_F + \|\mathcal{U}^{k}\|_F}{\mu^k} \\ &\leq \frac{c_L}{\sqrt{\mu^k}} + \frac{c_L}{\sqrt{\rho\mu^k}} \to 0, \end{aligned} \tag{B.31}$$

and similarly,

$$\begin{aligned} &\left\|\Gamma^{-1}(\dot{\mathbf{X}}^{k+1}) - \mathcal{T}^{k+1}\right\|_F \\ &\leq \frac{\|\mathcal{V}^{k+1}\|_F + \|\mathcal{V}^{k}\|_F}{\mu^k} \\ &\leq \frac{c_N}{\sqrt{\mu^k}} + \frac{c_N}{\sqrt{\rho\mu^k}} \to 0. \end{aligned} \tag{B.32}$$

These estimates establish (B.3) and (B.4).

Together with (B.27), the vanishing primal residuals imply

$$\begin{aligned} \dot{\mathbf{Z}}^k &\to \dot{\mathbf{X}}^\star, \\ \mathcal{S}^k &\to \Gamma^{-1}(\dot{\mathbf{X}}^\star), \\ \mathcal{T}^k &\to \Gamma^{-1}(\dot{\mathbf{X}}^\star). \end{aligned} \tag{B.33}$$

Setting

$$\dot{\mathbf{Z}}^\star = \dot{\mathbf{X}}^\star, \qquad \mathcal{S}^\star = \Gamma^{-1}(\dot{\mathbf{X}}^\star), \qquad \mathcal{T}^\star = \Gamma^{-1}(\dot{\mathbf{X}}^\star),$$

we conclude that all four primal variable sequences are Cauchy and converge to their respective limits. The estimates above hold for every realization of the sampled norm-preserving transformations. This completes the proof. □